

Online Resource 4: Architecture Ablation

Supporting Information for “When Inverse Physics-Informed Neural Networks Become Practically Identifiable: A Case Study in Synaptic Dopamine Transport”

Emmanuel Hackman and Huiqing Zhu
School of Mathematics and Natural Sciences,
University of Southern Mississippi, Hattiesburg, MS 39406, USA

This resource reports the architecture ablation study referenced in the Results section of the main text.

Architecture Ablation

To verify that the canonical four-hidden-layer, 64-neuron architecture is not badly over- or under-parameterized, we re-trained the forward problem with three MLP configurations on identical data and hyperparameters.

Table 1: **Architecture ablation.** Forward-problem L^2 error against the finite-difference reference and total training time (Adam + L-BFGS, single seed 1234) for three MLP configurations.

Configuration	Params	L^2 vs FD (%)	Train time (s)
[3, 32, 32, 32, 1] (compact)	2,273	1.38	241.1
[3, 64, 64, 64, 64, 1] (canonical)	12,801	1.52	329.5
[3, 128, 128, 128, 1] (wide)	33,665	2.04	308.6

All three configurations achieve sub-2.1% forward L^2 error; the wide configuration is slightly less accurate (2.04%), likely reflecting mild overparameterization for this smooth 2D problem. The canonical architecture is competitive with both alternatives and robust to moderate changes in capacity.